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Chassis top cover support integrated riser cage assembly

Abstract

An information handling system includes a connection tab physically coupled to a top cover of the information handling system, and a sliding bracket configured to engage with a riser cage of the information handling system and the connection tab of the top cover. A mezzanine card holder has an alignment sleeve through which a standoff of a system board of the information handling system may be inserted, such that the standoff may engage with the sliding bracket.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to information handling systems, and more particularly relates to a chassis top cover support integrated riser cage assembly.

BACKGROUND

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option is an information handling system. An information handling system generally processes, compiles, stores, or communicates information or data for business, personal, or other purposes. Technology and information handling needs and requirements can vary between different applications. Thus, information handling systems can also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information can be processed, stored, or communicated. The variations in information handling systems allow information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems can include a variety of

hardware and software resources that can be configured to process, store, and communicate information and can include one or more computer systems, graphics interface systems, data storage systems, networking systems, and mobile communication systems. Information handling systems can also implement various virtualized architectures. Data and voice communications among information handling systems may be via networks that are wired, wireless, or some combination.

SUMMARY

(3) An information handling system includes a connection tab physically coupled to a top cover of the information handling system, and a sliding bracket configured to engage with a riser cage of the information handling system and the connection tab of the top cover. A mezzanine card holder has an alignment sleeve through which a standoff of a system board of the information handling system may be inserted, such that the standoff may engage with the sliding bracket.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:
- (2) FIG. 1 is a block diagram illustrating an information handling system according to an embodiment of the present disclosure;
- (3) FIG. 2 is a perspective view of an information handling system according to an embodiment of the present disclosure;
- (4) FIG. 3 is a perspective view of a top cover of an information handling system, according to an embodiment of the present disclosure;
- (5) FIG. 4 is a perspective view of a section of an information handling system, according to an embodiment of the present disclosure;
- (6) FIG. 5 is a cross-sectional view taken along line A-A in FIG. 4;
- (7) FIG. 6 is a exploded view of a riser cage, according to an embodiment of the present disclosure;
- (8) FIG. 7 is a perspective view of a mezzanine card holder, according to an embodiment of the present disclosure;
- (9) FIG. 8 is a perspective view of a sliding bracket, according to an embodiment of the present disclosure;
- (10) FIG. 9 is a perspective view of a sliding bracket and a mezzanine card holder with a riser cage, according to an embodiment of the present disclosure;
- (11) FIG. 10 is a perspective view of a sliding bracket and a mezzanine card holder coupled to a stationary guide associated with a chassis of an information handling system, according to an embodiment of the present disclosure;
- (12) FIG. 11 is a perspective view of a mezzanine card holder and a section of a riser cage, according to an embodiment of the present disclosure;
- (13) FIG. 12 is a perspective view of a mezzanine card holder and a section of a riser cage, according to an embodiment of the present disclosure;
- (14) FIG. 13 is a view of a section of a riser cage and a sliding bracket in a locked position, according to an embodiment of the present disclosure;
- (15) FIG. 14 is a perspective view of a section of a riser cage and a sliding bracket in a locked position, according to an embodiment of the present disclosure;
- (16) FIG. 15 is a cross-sectional view of a sliding bracket coupled with a connection tab of a top cover and a standoff of a system board of an information handling system, according to an

embodiment of the present disclosure;

(17) FIG. **16** is a view of a section of a riser cage and a sliding bracket in an unlocked position, according to an embodiment of the present disclosure;

(18) FIG. **17** is a perspective view of a section of a riser cage and a sliding bracket in an unlocked position, according to an embodiment of the present disclosure; and

(19) FIG. **18** is a cross-sectional view of a sliding bracket coupled with a connection tab of a top cover and a standoff of a system board of an information handling system, according to an embodiment of the present disclosure.

(20) The use of the same reference symbols in different drawings indicates similar or identical items.

DETAILED DESCRIPTION OF THE DRAWINGS

(21) The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.

(22) FIG. **1** illustrates an embodiment of an information handling system **100** including processors **102** and **104**, a chipset **110**, a memory **120**, a graphics adapter **130** connected to a video display **134**, a non-volatile RAM (NV-RAM) **140** that includes a basic input and output system/extensible firmware interface (BIOS/EFI) module **142**, a disk controller **150**, a hard disk drive (HDD) **154**, an optical disk drive **156**, a disk emulator **160** connected to a solid-state drive (SSD) **164**, an input/output (I/O) interface **170** connected to an add-on resource **174** and a trusted platform module (TPM) **176**, a network interface **180**, and a baseboard management controller (BMC) **190**.

Processor **102** is connected to chipset **110** via processor interface **106**, and processor **104** is connected to the chipset via processor interface **108**. In a particular embodiment, processors **102** and **104** are connected together via a high-capacity coherent fabric, such as a HyperTransport link, a QuickPath Interconnect, or the like. Chipset **110** represents an integrated circuit or group of integrated circuits that manage the data flow between processors **102** and **104** and the other elements of information handling system **100**. In a particular embodiment, chipset **110** represents a pair of integrated circuits, such as a northbridge component and a southbridge component. In another embodiment, some or all of the functions and features of chipset **110** are integrated with one or more of processors **102** and **104**.

(23) Memory **120** is connected to chipset **110** via a memory interface **122**. An example of memory interface **122** includes a double data rate (DDR) memory channel and memory **120** represents one or more DDR dual in-line memory modules (DIMMs). In a particular embodiment, memory interface **122** represents two or more DDR channels. In another embodiment, one or more of processors **102** and **104** include a memory interface that provides a dedicated memory for the processors. A DDR channel and the connected DDR DIMMs can be in accordance with a particular DDR standard, such as a DDR3 standard, a DDR4 standard, a DDR5 standard, or the like.

(24) Memory **120** may further represent various combinations of memory types, such as dynamic random access memory (DRAM) DIMMs, static random access memory (SRAM) DIMMs, non-volatile DIMMs (NV-DIMMs), storage class memory devices, Read-Only Memory (ROM) devices, or the like. Graphics adapter **130** is connected to chipset **110** via a graphics interface **132** and provides a video display output **136** to a video display **134**. An example of a graphics interface **132** includes a PCIe interface and graphics adapter **130** can include a four-lane (×4) PCIe adapter, an eight-lane (×8) PCIe adapter, a 16-lane (×16) PCIe adapter, or another configuration, as needed or desired. In a particular embodiment, graphics adapter **130** is provided down on a system printed circuit board (PCB). Video display output **136** can include a Digital Video Interface (DVI), a High-Definition Multimedia Interface (HDMI), a DisplayPort interface, or the like, and video display **134** can include a monitor, a smart television, an embedded display such as a laptop computer

display, or the like.

(25) NV-RAM **140**, disk controller **150**, and I/O interface **170** are connected to chipset **110** via an I/O channel **112**. An example of I/O channel **112** includes one or more point-to-point PCIe links between chipset **110** and each of NV-RAM **140**, disk controller **150**, and I/O interface **170**.

Chipset **110** can also include one or more other I/O interfaces, including a PCIe interface, an Industry Standard Architecture (ISA) interface, a Small Computer Serial Interface (SCSI) interface, an Inter-Integrated Circuit (I2C) interface, a Serial Peripheral Interface (SPI), a Universal Serial Bus (USB), another interface, or a combination thereof. NV-RAM **140** includes BIOS/EFI module **142** that stores machine-executable code (BIOS/EFI code) that operates to detect the resources of information handling system **100**, to provide drivers for the resources, to initialize the resources, and to provide common access mechanisms for the resources. The functions and features of BIOS/EFI module **142** will be further described below.

(26) Disk controller **150** includes a disk interface **152** that connects the disk controller to a hard disk drive (HDD) **154**, to an optical disk drive (ODD) **156**, and to disk emulator **160**. An example of disk interface **152** includes an Integrated Drive Electronics (IDE) interface, an Advanced Technology Attachment (ATA) such as a parallel ATA (PATA) interface or a serial ATA (SATA) interface, a SCSI interface, a USB interface, a proprietary interface, or a combination thereof. Disk emulator **160** permits SSD **164** to be connected to information handling system **100** via an external interface **162**. An example of external interface **162** includes a USB interface, an Institute of Electrical and Electronics Engineers (IEEE) 1394 (Firewire) interface, a proprietary interface, or a combination thereof. Alternatively, SSD **164** can be disposed within information handling system **100**.

(27) I/O interface **170** includes a peripheral interface **172** that connects the I/O interface to add-on resource **174**, to TPM **176**, and to network interface **180**. Peripheral interface **172** can be the same type of interface as I/O channel **112** or can be a different type of interface. As such, I/O interface **170** extends the capacity of I/O channel **112** when peripheral interface **172** and the I/O channel are of the same type, and the I/O interface translates information from a format suitable to the I/O channel to a format suitable to the peripheral interface **172** when they are of a different type. Add-on resource **174** can include a data storage system, an additional graphics interface, a network interface card (NIC), a sound/video processing card, another add-on resource, or a combination thereof. Add-on resource **174** can be on a main circuit board, on separate circuit board, or add-in card disposed within information handling system **100**, a device that is external to the information handling system, or a combination thereof.

(28) Network interface **180** represents a network communication device disposed within information handling system **100**, on a main circuit board of the information handling system, integrated onto another component such as chipset **110**, in another suitable location, or a combination thereof. Network interface **180** includes a network channel **182** that provides an interface to devices that are external to information handling system **100**. In a particular embodiment, network channel **182** is of a different type than peripheral interface **172**, and network interface **180** translates information from a format suitable to the peripheral channel to a format suitable to external devices.

(29) In a particular embodiment, network interface **180** includes a NIC or host bus adapter (HBA), and an example of network channel **182** includes an InfiniBand channel, a Fibre Channel, a Gigabit Ethernet channel, a proprietary channel architecture, or a combination thereof. In another embodiment, network interface **180** includes a wireless communication interface, and network channel **182** includes a Wi-Fi channel, a near-field communication (NFC) channel, a Bluetooth® or Bluetooth-Low-Energy (BLE) channel, a cellular based interface such as a Global System for Mobile (GSM) interface, a Code-Division Multiple Access (CDMA) interface, a Universal Mobile Telecommunications System (UMTS) interface, a Long-Term Evolution (LTE) interface, or another cellular based interface, or a combination thereof. Network channel **182** can be connected to an

external network resource (not illustrated). The network resource can include another information handling system, a data storage system, another network, a grid management system, another suitable resource, or a combination thereof.

(30) BMC **190** is connected to multiple elements of information handling system **100** via one or more management interface **192** to provide out of band monitoring, maintenance, and control of the elements of the information handling system. As such, BMC **190** represents a processing device different from processor **102** and processor **104**, which provides various management functions for information handling system **100**. For example, BMC **190** may be responsible for power management, cooling management, and the like. The term BMC is often used in the context of server systems, while in a consumer-level device, a BMC may be referred to as an embedded controller (EC). A BMC included in a data storage system can be referred to as a storage enclosure processor. A BMC included at a chassis of a blade server can be referred to as a chassis management controller and embedded controllers included at the blades of the blade server can be referred to as blade management controllers. Capabilities and functions provided by BMC **190** can vary considerably based on the type of information handling system. BMC **190** can operate in accordance with an Intelligent Platform Management Interface (IPMI). Examples of BMC **190** include an Integrated Dell R Remote Access Controller (iDRAC).

(31) Management interface **192** represents one or more out-of-band communication interfaces between BMC **190** and the elements of information handling system **100**, and can include a I2C bus, a System Management Bus (SMBus), a Power Management Bus (PMBUS), a Low Pin Count (LPC) interface, a serial bus such as a Universal Serial Bus (USB) or a Serial Peripheral Interface (SPI), a network interface such as an Ethernet interface, a high-speed serial data link such as a PCIe interface, a Network Controller Sideband Interface (NC-SI), or the like. As used herein, out-of-band access refers to operations performed apart from a BIOS/operating system execution environment on information handling system **100**, that is apart from the execution of code by processors **102** and **104** and procedures that are implemented on the information handling system in response to the executed code.

(32) BMC **190** operates to monitor and maintain system firmware, such as code stored in BIOS/EFI module **142**, option ROMs for graphics adapter **130**, disk controller **150**, add-on resource **174**, network interface **180**, or other elements of information handling system **100**, as needed or desired. In particular, BMC **190** includes a network interface **194** that can be connected to a remote management system to receive firmware updates, as needed or desired. BMC **190** receives the firmware updates, stores the updates to a data storage device associated with the BMC, transfers the firmware updates to NV-RAM of the device or system that is the subject of the firmware update, thereby replacing the currently operating firmware associated with the device or system, and reboots information handling system, whereupon the device or system utilizes the updated firmware image.

(33) BMC **190** utilizes various protocols and application programming interfaces (APIs) to direct and control the processes for monitoring and maintaining the system firmware. An example of a protocol or API for monitoring and maintaining the system firmware includes a graphical user interface (GUI) associated with BMC **190**, an interface defined by the Distributed Management Taskforce (DMTF) (such as a Web Services Management (WSMan) interface, a Management Component Transport Protocol (MCTP) or, a RedfishR interface), various vendor defined interfaces (such as a Dell EMC Remote Access Controller Administrator (RACADM) utility, a Dell EMC OpenManage Enterprise, a Dell EMC OpenManage Server Administrator (OMSS) utility, a Dell EMC OpenManage Storage Services (OMSS) utility, or a Dell EMC OpenManage Deployment Toolkit (DTK) suite), a BIOS setup utility such as invoked by a “F2” boot option, or another protocol or API, as needed or desired.

(34) In a particular embodiment, BMC **190** is included on a main circuit board (such as a baseboard, a motherboard, or any combination thereof) of information handling system **100** or is

integrated onto another element of the information handling system such as chipset **110**, or another suitable element, as needed or desired. As such, BMC **190** can be part of an integrated circuit or a chipset within information handling system **100**. An example of BMC **190** includes an iDRAC, or the like. BMC **190** may operate on a separate power plane from other resources in information handling system **100**. Thus BMC **190** can communicate with the management system via network interface **194** while the resources of information handling system **100** are powered off. Here, information can be sent from the management system to BMC **190** and the information can be stored in a RAM or NV-RAM associated with the BMC. Information stored in the RAM may be lost after power-down of the power plane for BMC **190**, while information stored in the NV-RAM may be saved through a power-down/power-up cycle of the power plane for the BMC.

(35) Information handling system **100** can include additional components and additional busses, not shown for clarity. For example, information handling system **100** can include multiple processor cores, audio devices, and the like. While a particular arrangement of bus technologies and interconnections is illustrated for the purpose of example, one of skill will appreciate that the techniques disclosed herein are applicable to other system architectures. Information handling system **100** can include multiple central processing units (CPUs) and redundant bus controllers. One or more components can be integrated together. Information handling system **100** can include additional buses and bus protocols, for example, I2C and the like. Additional components of information handling system **100** can include one or more storage devices that can store machine-executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display.

(36) For purposes of this disclosure information handling system **100** can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, information handling system **100** can be a personal computer, a laptop computer, a smartphone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch, a router, or another network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Further, information handling system **100** can include processing resources for executing machine-executable code, such as processor **102**, a programmable logic array (PLA), an embedded device such as a System-on-a-Chip (SoC), or other control logic hardware. Information handling system **100** can also include one or more computer-readable media for storing machine-executable code, such as software or data.

(37) An information handling system such as a rackmount server, with a size of two rack units, is typically designed with a long-span top cover with minimal if any support. This is because typically there is no space to add a support feature, also referred to simply as a support, for the top cover that is integrated with a chassis of the information handling system because the information handling system is usually fully occupied by various components, such as a motherboard, riser cages, Peripheral Component Interconnect-Express (PCIe) cards, etc. An issue arises when using a vacuum hoist lift to transport the information handling system. Because the vacuum hoist lift typically grips and lifts the information handling system via its top cover, the weight of the information handling system is held by the top cover. Because of its long span, when there is inadequate support for the top cover, the top cover may be deformed during transport. As the weight of the information handling system increases, the chance that the top cover deforms also increases. This may result in additional issues. For example, sliding the information handling system into a rack chassis may be difficult due to limited margins allotted for each slot. Thus, there is a need for a support for the top cover with a better structure integration within the chassis of the information handling system. This allows transporting the information handling system using the top cover, such as by the vacuum hoist lift, without the risk of deforming the top cover.

(38) FIG. 2 shows a perspective view of an information handling system **200**. Information handling system **200**, which is similar to information handling system **100** of FIG. 1, maybe a rackmount server that includes a chassis **205** and riser cages **210** and **215**. Information handling system **200** also includes a system board **225**, also referred to herein as a motherboard, a portion of which is disposed underneath riser cages **210** and **215**. Information handling system **200** is shown with a top cover for chassis **205** in transparent view to highlight features of embodiments of the present disclosure.

(39) In this example, the present disclosure includes a top cover support assembly with re-designed top cover, riser cage, and mezzanine card holder components. The top cover support also includes a sliding bracket to engage with the top cover, riser cage, and mezzanine card holder, securing the top cover and motherboard support tray as one solid assembly part. For example, the top cover includes a connection tab feature, also referred to simply as a connection tab, which may be configured to engage with a connection slot of the riser cage, also referred to herein as a riser cage slot or simply as a slot. The connection slot may also be configured to be aligned with another connection slot of a sliding bracket.

(40) The sliding bracket may also be configured with a U-shaped notch or channel that is aligned with a similar U-shaped notch or channel of the connection tab and an alignment sleeve of the mezzanine card holder. The alignment sleeve of the mezzanine cardholder may be configured through which a standoff of support tray of a system board, also referred to as a motherboard, of the information handling system, may be inserted. The support tray of the system board may be physically coupled to the information handling system's chassis. Thus, providing support for the top cover of information handling system **200**. One of ordinary skill in the art will appreciate that an information handling device in accordance with embodiments of the present disclosure may include additional components than those illustrated in the figures.

(41) Information handling system **200** may be any size that is configured to fit within a chassis, enclosure, or server rack. For example, information handling system **200** may be of any size, including, but not limited to, a one rack unit of height rackmount server or a two rack unit rackmount server, etc. Moreover, information handling system **200** may represent a full-width rackmount server, a half-width rackmount server, or any other width rackmount server. Information handling system **100** includes a section **220** which is shown in greater detail in FIG. 2.

(42) Riser cages **210** and **215**, also referred to herein simply as risers, may be used to extend a slot for a chip or card in a fully loaded information handling system. For example, riser cages **210** and **215** may secure Peripheral Component Interconnect (PCI) cards, PCIe cards, mezzanine cards, or the like. Further, riser cages **210** and **215** may be butterfly riser cages. In one embodiment, riser cages **210** and **215** may latch to a chassis of information handling system **200**. One of ordinary skill in the art will appreciate that riser cages **210** and **215** may be of varying length and width and may retain PCI, PCIe, and/or mezzanine cards of varying length and/or width. Each of riser cages **210** and **215** includes features discussed in greater detail below. For example, although the illustrations refer to riser cage **210**, riser cage **215** may include a mezzanine card holder and a sliding bracket configured to provide top cover support that is integrated into the chassis of information handling system **200**.

(43) FIG. 3 shows a top cover **300** of information handling system **200** of FIG. 2. Top cover **300** includes connection tabs **305** and **310**. Connection tabs **305** and **310** may be t-connection tabs, such that each connection tab may engage with a connection slot of a riser cage. For example, connection tab **305** may be configured to engage with a connection slot of riser cage **210** and its support. Connection tab **310** may be configured to engage with a connection slot of riser cage **215** and its support. Top cover **300** may be pivotally, slidably, or otherwise removably attached to information handling system **200** of FIG. 2. Connection tabs **305** and **310** may be of suitable material which may be similar to the material used for top cover **300**.

(44) FIG. 4 shows an expanded view of section **220** of information handling system **200** of FIG. 2

in greater detail. The expanded view is shown with a top cover of information handling system **200** set to transparent except for connection tabs **305** and **310**. This is performed to highlight features of embodiments of the present disclosure. Section **220** shows a connection slot **420** of riser cage **210**. Connection slot **420** may be configured to align with a connection tab of a sliding bracket **405** as shown in greater detail below. The connection slots may be used to physically couple riser cage **210** and sliding bracket **405** to a connection tab of a top cover of information handling system **200**. Accordingly, riser cage **215** includes a connection slot **430** that may be configured to align with a connection slot of sliding bracket **405**.

(45) The connection slots may be used to physically couple riser cage **215** and sliding bracket **405** to a connection tab of the top cover of information handling system **200**. For example, connection tab **305** may be configured to physically couple with an undercut or U-shaped notch of connection slot **420**. Similarly, connection tab **310** may be configured to physically couple with an undercut or U-shaped notch of connection slot **420**. The expanded view also shows a mezzanine card holder **410** with an alignment sleeve **425**. Alignment sleeve **425** may be used to physically couple mezzanine card holder **410** with a standoff of a system board of information handling system **200** as shown in greater detail below. Sliding bracket **405** and its components along with mezzanine card holder **410** may be of suitable material, such as a non-ferromagnetic material like hard plastic or similar.

(46) FIG. 5 is a cross-sectional view **500** taken along line A-A as shown in FIG. 4. This view shows how sliding bracket **405** may be physically coupled to riser cage **210** via fasteners **505** and **510** through guide slots **530** and **535** respectively, wherein guide slots **530** may be elongated slots or grooves. Sliding bracket **405** may also be secured to riser cage **210** via fastener **550**. Examples of fasteners include rivets, screws, machine screws, bolts, nuts, pins, ties, etc. Fasteners **505**, **510**, and **550** may be mounting screws or similar. In particular, fastener **550** may be a plunger lock or similar. Further, this view shows a physical engagement of connection tab **305** to riser cage **210** and sliding bracket **405**. This physical engagement may allow the top cover of the information handling system to be supported, such as when a hoist vacuum is used to carry the information handling system from one location to another. Thus, avoiding the possibility of deforming the top cover, such as top cover **300** of FIG. 3.

(47) This view also shows a physical engagement of riser cage **210** and sliding bracket **405** to a system board **520** via mezzanine card holder **410** and a guiding rail **515**, wherein guiding rail **515** is disposed between sliding bracket **405** and mezzanine card holder **410**. A standoff **525** of a system board of the information handling system is disposed through an opening of alignment sleeve **425**. Standoff **525** may be a T-standoff which includes a head **540** and a base **545**. Head **540** may secure system board **520** to sliding bracket **405**. Base **545** may be threaded or unthreaded and physically coupled to a motherboard tray of a chassis of the information handling system through system board **520**. This allows sliding bracket **405** to be secured to system board **520** and/or the motherboard tray of the chassis of the information handling system.

(48) FIG. 6 shows an exploded view of riser cage **210**. Riser cage **210** includes a main body **610**, sliding bracket **405**, and mezzanine card holder **410** with alignment sleeve **425**. Main body **610** includes stationary guides **620** and **630** which may be used to secure sliding bracket **405** and mezzanine card holder **410** to main body **610**. The stationary guides may also be referred to as connection grooves. Sliding bracket **405** includes surfaces **625**, **635**, and **640**. Surface **625** may be disposed adjacent to one end of surface **640**. For example, surfaces **625** and **640** may be positioned at approximately 90 degrees relative to each other. Further, the length of surface **625** may be approximately the same as the length of surface **640**. Similarly, surface **635** may be disposed adjacent to another end of surface **640**. For example, surfaces **635** and **640** may be positioned at approximately 90 degrees relative to each other. Further, the length of surface **635** may be approximately three-quarters or so of surface **640**.

(49) FIG. 7 shows a perspective view of mezzanine card holder **410** with alignment sleeve **425**.

Alignment sleeve **425** includes an opening **715** with a diameter that is greater than standoff **525**. Mezzanine card holder **410** includes mounting holes, such as mounting holes **705** and **710** which may be used to physically couple mezzanine card holder **410** to riser cage **210** via one or more fasteners.

(50) FIG. **8** shows a perspective view of sliding bracket **405**. Sliding bracket **405** includes surfaces **640**, **635**, and **625** which are configured with different features. The view includes fastener **550** along one end of surface **640**. Surface **640** includes two guide slots **530** and **535** that are next to each other along surface **640**. In this example, guide slot **530** may be located around the middle of sliding bracket **405** between fastener **550** and guide slot **535**. Surface **640** may also include a slot for fastener **550** to rest into. The guide slots may be used to slide or glide sliding bracket **405** to physically couple with riser cage **210**.

(51) Surface **635** includes opening **415** which may be a semi-circular notch, U-shaped slot, or channel at one end of surface **635**. The semi-circular notch may be used to interlock with a standoff of the information handling system's system board when sliding bracket **405** is in a locked position. Approximately three-quarters or so of the length of surface **635** is thinner than the remaining length of surface **635**. Surface **625** may include connection slot **645**. Connection slot **645** may be aligned with connection slot **420** of FIG. **4**. Further connection slot **645** may include a semi-circular notch, U-shaped slot, or channel at one section. This notch may align with the notch at surface **635**. Surface **625** may also include rivet **805** which may be used to physically couple surfaces **635** and **625**, wherein a portion **810** of surface **625** may be folded perpendicularly to meet with an underside of surface **635**.

(52) FIG. **9** shows a perspective view **900** of a section of riser cage **210** and sliding bracket **405**. Perspective view **900** shows an opening **415** which may be aligned with opening **715** of alignment sleeve **425**. Sliding bracket **405** may be configured to engage with stationary guide **620** and **630**, such that connection slot **645** is aligned with connection slot **420**. Sliding bracket **405** may be secured in place by fasteners **505**, **510**, and **550**.

(53) FIG. **10** shows a perspective view **1000** which is similar to perspective view **900** of FIG. **9** with main body **610** of riser cage **210** removed to show additional details associated with the present disclosure. For example, perspective view **1000** shows that stationary guide **630** is disposed between sliding bracket **405** and mezzanine card holder **410**. Further, perspective view **1000** shows rivet **805** which may be used to physically couple surfaces **635** and **625**. Rivet **805** may be located above opening **715** of mezzanine card holder **410**. Perspective view **1000** also shows a location of connection slot **645** relative to rivet **805**.

(54) FIG. **11** shows a perspective view **1100** of mezzanine card holder **410** and a section of riser cage **210**. Perspective view **1100** shows mezzanine card holder **410** physically coupled to riser cage **210** via stationary guides **620** and **630** using one or more fasteners, such as fastener **1105**.

(55) FIG. **12** shows an exploded view **1200** of mezzanine card holder **410** and a section of riser cage **210**. Exploded view **1200** shows how mezzanine card holder **410** may be physically coupled to riser cage **210** via stationary guides **620** and **630**. For example, mezzanine card holder **410** may be coupled via one or more fasteners, such as fastener **1105**, to one or more mounting holes, such as mounting hole **1205**.

(56) FIG. **13** shows a perspective view **1300** of sliding bracket **405** and a section of riser cage **210**. Perspective view **1300** shows sliding bracket **405** in a locked position **1305**. Fastener **550** may be used to move sliding bracket **405** from locked position **1305** to unlocked position **1310** or vice versa. When in locked position **1305**, fastener **550** may rest inside a slot and/or rotated into the slot to keep sliding bracket **405** in locked position **1305** and disposed adjacent to a locked marker image. When sliding bracket **405** is in locked position **1305**, an opening **1315** may be formed along a channel of connection slot **420**, wherein a connection tab associated with the top cover of the information handling system may engage.

(57) FIG. **14** shows a perspective view **1400** of sliding bracket **405** and a section or riser cage **210**.

Perspective view **1400** shows sliding bracket **405** in locked position **1305**. When sliding bracket **405** is in locked position **1305**, fastener **550** may be positioned towards one end of guide slot **530** away from fastener **550**. In addition, when in the locked position, connection slot **420** may be aligned with connection slot **645**.

(58) FIG. **15** shows a cross-section view **1500** of sliding bracket **405** engaged with connection tab **305** of top cover **300**. Perspective view **1400** shows how connection tab **305** of top cover **300** may be engaged with connection slot **420** and guide slot **530**. Perspective view **1400** also shows a position of connection tab **305** relative to fastener **505**.

(59) FIG. **16** shows a perspective view **1600** of sliding bracket **405** and a section of riser cage **210**. Perspective view **1600** shows sliding bracket **405** in unlocked position **1310**. When in unlocked position **1310**, fastener **550** may rest outside a slot associated with locked position **1305** and disposed adjacent to an unlock marker image. When sliding bracket **405** is in unlocked position **1310**, connection slot **420** may not be aligned with connection slot **645**, as such opening **1315** may be closed or not viable as depicted by encircled section **1605**.

(60) FIG. **17** shows a perspective view **1700** of sliding bracket **405** and a section of riser cage **210**. Perspective view **1700** shows sliding bracket **405** in unlocked position **1310**. When sliding bracket **405** is in unlocked position **1310**, fastener **550** may be positioned towards another end of guide slot **530** towards fastener **550**. In addition, when in an unlocked position, connection slot **420** may not be aligned with connection slot **645** as depicted by encircled section **1705**.

(61) FIG. **18** shows a cross-section view **1800** of sliding bracket **405** and disengagement of connection tab **305** of top cover **300**. An encircled section **1805** of cross-section view **1800** shows how connection tab **305** may hit fastener **510** when a user tries to engage top cover **300** when sliding bracket **405** is in unlocked position **1310**. Thus, the top cover support may not be engaged or assembled in place because the connection tab of the top cover may be blocked by a surface of sliding bracket **405**.

(62) Those of ordinary skill in the art will appreciate that the configuration and/or hardware components of the top cover support of the information handling system depicted above. For example, the illustrative components are not intended to be exhaustive but rather are representative to highlight components that can be utilized to implement aspects of the present disclosure. For example, other devices and/or components may be used in addition to or in place of the devices/components depicted. The depicted example does not convey or imply any architectural or other limitations with respect to the presently described embodiments and/or the general disclosure. In the discussion of the figures, reference may also be made to components illustrated in other figures for continuity of the description.

(63) When referred to as a “device,” a “module,” a “unit,” a “controller,” or the like, the embodiments described herein can be configured as hardware. For example, a portion of an information handling system device may be hardware such as, for example, an integrated circuit (such as an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a structured ASIC, or a device embedded in a larger chip), a card (such as a Peripheral Component Interface (PCI) card, a PCI-express card, a Personal Computer Memory Card International Association (PCMCIA) card, or other such expansion card), or a system (such as a motherboard, a system-on-a-chip (SoC), or a stand-alone device).

(64) The present disclosure contemplates a computer-readable medium that includes instructions or receives and executes instructions responsive to a propagated signal; so that a device connected to a network can communicate voice, video, or data over the network. Further, the instructions may be transmitted or received over the network via the network interface device.

(65) While the computer-readable medium is shown to be a single medium, the term “computer-readable medium” includes a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of instructions. The term “computer-readable medium” shall also include any medium that is capable of storing, encoding or

carrying a set of instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein.

(66) In a particular non-limiting, exemplary embodiment, the computer-readable medium can include a solid-state memory such as a memory card or other package that houses one or more non-volatile read-only memories. Further, the computer-readable medium can be a random-access memory or other volatile re-writable memory. Additionally, the computer-readable medium can include a magneto-optical or optical medium, such as a disk or tapes, or another storage device to store information received via carrier wave signals such as a signal communicated over a transmission medium. A digital file attachment to an e-mail or other self-contained information archive or set of archives may be considered a distribution medium that is equivalent to a tangible storage medium. Accordingly, the disclosure is considered to include any one or more of a computer-readable medium or a distribution medium and other equivalents and successor media, in which data or instructions may be stored.

(67) Although only a few exemplary embodiments have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the embodiments of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the embodiments of the present disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents but also equivalent structures.

Claims

1. A support for a top cover of an information handling system, the support comprising: a connection tab physically coupled to the top cover of the information handling system; a sliding bracket configured to engage with a riser cage of the information handling system and the connection tab of the top cover; and a mezzanine card holder with an alignment sleeve through which a standoff of a system board of the information handling system may be inserted, wherein the standoff may engage with the sliding bracket.
2. The support of claim 1, wherein the sliding bracket may slide along a stationary guide of the riser cage.
3. The support of claim 1, wherein the sliding bracket includes a first connection slot used to engage with the connection tab of the top cover, wherein the first connection slot is configured to align with a second connection slot of the riser cage.
4. The support of claim 3, wherein the first connection slot and the second connection slot are aligned when the sliding bracket is in a locked position.
5. The support of claim 3, wherein the first connection slot and the second connection slot are not aligned when the sliding bracket is in an unlocked position.
6. The support of claim 1, wherein the sliding bracket includes a channel configured to align with the alignment sleeve.
7. The support of claim 1, wherein the sliding bracket includes a plunger lock which is configured to engage the sliding bracket in a locked position.
8. The support of claim 1, wherein the standoff is further physically coupled to a support tray of the system board.
9. The support of claim 1, wherein the mezzanine card holder is mounted to the riser cage.
10. An information handling system, comprising: a connection tab physically coupled to a top cover of the information handling system; a sliding bracket configured to engage with a riser cage of the information handling system and the connection tab of the top cover; and a mezzanine card holder with an alignment sleeve through which a standoff of a system board of the information

handling system may be inserted, wherein the standoff may engage with the sliding bracket.

11. The information handling system of claim 10, wherein the sliding bracket may slide along a stationary guide of the riser cage.

12. The information handling system of claim 10, wherein the sliding bracket includes a first connection slot used to engage with the connection tab of the top cover, wherein the first connection slot is configured to align with a second connection slot of the riser cage.

13. The information handling system of claim 12, wherein the first connection slot and the second connection slot are aligned when the sliding bracket is in a locked position.

14. The information handling system of claim 12, wherein the first connection slot and the second connection slot are not aligned when the sliding bracket is in an unlocked position.

15. The information handling system of claim 10, wherein the sliding bracket includes a channel configured to align with the alignment sleeve.

16. An information handling system comprising: a connection tab physically coupled to a top cover of the information handling system; a sliding bracket configured to engage with a riser cage via a stationary guide of the riser cage, the sliding bracket further configured to engage the connection tab of the top cover; and a mezzanine card holder mounted to the riser cage, wherein the mezzanine card holder includes an alignment sleeve through which a standoff of a system board of the information handling system may be inserted.

17. The information handling system of claim 16, wherein the standoff is further physically coupled to a support tray of the system board.

18. The information handling system of claim 16, wherein the sliding bracket includes a first connection slot used to engage with the connection tab of the top cover, wherein the first connection slot is configured to align with a second connection slot of the riser cage.

19. The information handling system of claim 18, wherein the first connection slot and the second connection slot are aligned when the sliding bracket is in a locked position.

20. The information handling system of claim 18, wherein the first connection slot and the second connection slot are not aligned when the sliding bracket is in an unlocked position.
